Question 2: let $f: \Omega \to \Omega'$ be a function and let $\mathcal{F}$ be a ~~collection of subset~~ σ-field in Ω'. show that $f^{-1}(\mathcal{F}) := \{f^{-1}(A) : A \in \mathcal{F}\}$ is a σ-field in Ω. (10)

Solution:

(i) suppose $\phi \in \mathcal{F}$

$\Rightarrow f^{-1}(\phi) = \phi \in \mathcal{F}_1$

what is $\mathcal{F}_1$?

$\mathcal{F}_1 = f^{-1}(\mathcal{F})$.

(ii) suppose $A \in \mathcal{F}$

Need to show, $A^c \in \mathcal{F}$

$f^{-1}(A^c) \in \mathcal{F}_1$

$\Rightarrow (f^{-1}(A))^c \in \mathcal{F}_1$

suppose $\omega_1 \in f^{-1}(A^c)$

$\Rightarrow f(\omega_1) \in A^c$

$\Rightarrow f(\omega_1) \notin A$

$\Rightarrow \omega_1 \notin f^{-1}(A)$

$\Rightarrow \omega_1 \in (f^{-1}(A))^c$

so that $f^{-1}(F)$ is closed under complementation.

(iii) suppose $A_1, A_2, \ldots \in \mathcal{F}$

Need to check that $\bigcup_{n=1}^{\infty} A_n \in \mathcal{F}$

$f^{-1}\left(\bigcup_{n=1}^{\infty} A_n\right) \in \mathcal{F}_1$

$\Rightarrow f^{-1}(A_1 \cup A_2 \cup \ldots \cup A_k \cup \ldots) \in \mathcal{F}_1$

$\Rightarrow f^{-1}(A_1) \cup f^{-1}(A_2) \ldots \cup f^{-1}(A_k) \cup \ldots \in \mathcal{F}_1$

$\Rightarrow \bigcup_{n=1}^{\infty} f^{-1}(A_n) \in \mathcal{F}_1$ (because this follows from the property of Inverse function)

~~so t~~

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Therefore $f^{-1}(F)$ is closed under countable union.

Hence $\phi \in f^{-1}(F)$, $f^{-1}(F)$ is closed under complementation and countable union also. It satisfies all the properties of a σ-field. so that $f^{-1}(F) := \{f^{-1}(A) / A \in F\}$ is a σ-field in Ω.

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Question:- 3:- Let $f: \Omega \to \Omega'$ be a function and let $\mathcal{F}$ be a collection of subset of Ω'. Show that:-

$$\sigma(f^{-1}(\mathcal{F})) = f^{-1}(\sigma(\mathcal{F}))$$

where, $f^{-1}(A)$ denotes the pre-image of $A \subset \Omega'$;

$f^{-1}(\mathcal{F}) = \{f^{-1}(A) : A \in \mathcal{F}\}$ and

$f^{-1}(\sigma(\mathcal{F})) = \{f^{-1}(A) ; A \in \sigma(\mathcal{F})\}$.

Answer:- We know that:-

Result:- 1⇒ if $\mathcal{F}$ is a σ-field in Ω', then $f^{-1}(\mathcal{F})$ is a σ-field in Ω. // Problem 2 | Can use directly

Proof:- (i) $\Omega' \in \mathcal{F}$ we have

$$f^{-1}(\Omega') = \Omega \in f^{-1}(\mathcal{F})$$

by, (i) inverse map property which say

$$f^{-1}(\Omega') = \Omega.$$

(ii) If $A \in \mathcal{F}$ then $A^c \in \mathcal{F}$, and so if $f^{-1}(A) \in f^{-1}(\mathcal{F})$, we have

$$f^{-1}(A^c) = (f^{-1}(A))^c \in f^{-1}(\mathcal{F})$$

by (iii) inverse map property. $f^{-1}(A^c) = (f^{-1}(A))^c$

Hence, $f^{-1}(\mathcal{F})$ closed under complimentation.

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(iii) If $f^{-1}(\mathcal{F}_n) \in f^{-1}(\mathcal{F})$

then, $\bigcup_{n=1}^{\infty} f^{-1}(\mathcal{F}_n) = f^{-1}\left(\bigcup_n \mathcal{F}_n\right) \in f^{-1}(\mathcal{F})$

Since, $\bigcup_n \mathcal{F}_n \in \mathcal{F}$

$\Rightarrow$ $f^{-1}[\mathcal{F}]$ closed under countable union.

Hence, $f^{-1}(\mathcal{F})$ is a σ-field.

Now, by Result :-1 if $\sigma(\mathcal{F})$ is a σ-field in Ω' then $f^{-1}(\sigma(\mathcal{F}))$ is σ-field in Ω.

So, $f^{-1}(\sigma(\mathcal{F})) \supset f^{-1}(\mathcal{F})$

Since, $\sigma(\mathcal{F}) \supset \mathcal{F}$ and hence, by minimality,

$$f^{-1}(\sigma(\mathcal{F})) \supset \sigma(f^{-1}(\mathcal{F})) \quad \text{---- (*)}$$

Define:-

$$\mathcal{C}_e = \{\mathcal{F}' \in \Omega' : f^{-1}[\mathcal{F}') \in \sigma(f^{-1}(\mathcal{F}))\}$$

Then, $\mathcal{C}_e$ is a σ-field

Proof:- (i) $\Omega' \in \mathcal{C}_e$, since, $f^{-1}(\Omega') = \Omega \in \sigma(f^{-1}(\mathcal{F}))$

(ii) $A \in \mathcal{C}_e$ implies $A^c \in \mathcal{C}_e$

Since, $f^{-1}(A^c) = (f^{-1}(A))^c \in \sigma(f^{-1}(\mathcal{F}))$

if $f^{-1}(A) \in \sigma(f^{-1}(\mathcal{F}))$

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(iii) $\mathcal{F}_n' \in \mathcal{C}_e$ implies $\bigcup_n \mathcal{F}_n' \in \mathcal{C}_e$

Since,

$$f^{-1}\left(\bigcup_n \mathcal{F}_n'\right) = \bigcup_n f^{-1}(\mathcal{F}_n') \in \sigma(f^{-1}(\mathcal{F}))$$

if $f^{-1}(\mathcal{F}_n') \in \sigma(f^{-1}(\mathcal{F}))$

by defination,

~~$f^{-1}(\mathcal{C}_e) \in \sigma(f^{-1}(\mathcal{F}))$~~

~~Also,~~

~~$\mathcal{F} \in$~~

$$f^{-1}(\mathcal{C}_e) \subset \sigma(f^{-1}(\mathcal{F})) \quad \ldots (1)$$

Also,

$$\mathcal{F} \subset \mathcal{C}_e$$

Since, $f^{-1}(\mathcal{F}) \subset \sigma(f^{-1}(\mathcal{F}))$

Also, $\mathcal{C}_e$ is a σ-field

$\Rightarrow$

$$\sigma(\mathcal{F}) \subset \mathcal{C}_e \quad \ldots (2)$$

by (1) and (2)

$$f^{-1}(\sigma(\mathcal{F})) \subset f^{-1}(\mathcal{C}_e) \subset \sigma(f^{-1}(\mathcal{F}))$$

$\Rightarrow$

$$f^{-1}(\sigma(\mathcal{F})) \subset \sigma(f^{-1}(\mathcal{F})) \quad \ldots (**)$$

from (*) and (**)

$$f^{-1}(\sigma(\mathcal{F})) = \sigma(f^{-1}(\mathcal{F}))$$

Hence proof

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5 Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. Fix $A \in \mathcal{F}$ show that

$$\mathcal{C} = \{ B \in \mathcal{F} : \mathbb{P}(A \cap B) = \mathbb{P}(A) \cdot \mathbb{P}(B) \}$$ is a monotone class.

Solution

Here we want to prove that $\mathcal{C} = \{ B \in \mathcal{F} : \mathbb{P}(A \cap B) = \mathbb{P}(A) \cdot \mathbb{P}(B) \}$ is a monotone class. for that take $\{B_n\}$ be an increasing sequence from $\mathcal{C}$.

$\therefore \mathbb{P}(A \cap B_n) = \mathbb{P}(A) \cdot \mathbb{P}(B_n) \quad \forall n$

Now, $\mathbb{P}\left(A \cap \left(\bigcup_{n=1}^{\infty} B_n\right)\right)$

$= \mathbb{P}\left(\bigcup_{n=1}^{\infty} (A \cap B_n)\right)$

[$\{A \cap B_n\}$ is also an increasing sequence]

$= \lim_{n \to \infty} \mathbb{P}(A \cap B_n)$

$= \lim_{n \to \infty} \mathbb{P}(A) \cdot \mathbb{P}(B_n)$

$= \mathbb{P}(A) \cdot \lim_{n \to \infty} \mathbb{P}(B_n)$

$= \mathbb{P}(A) \cdot \mathbb{P}\left(\bigcup_{n=1}^{\infty} B_n\right)$

$\Rightarrow \bigcup_{n=1}^{\infty} B_n \in \mathcal{C}$

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Let $\{C_n\}$ be a decreasing sequence from $\mathcal{C}$

$\therefore \mathbb{P}(A \cap C_n) = \mathbb{P}(A) \cdot \mathbb{P}(C_n) \quad \forall n$

Now, $\mathbb{P}\left(A \cap \left(\bigcap_{n=1}^{\infty} C_n\right)\right)$

$= \mathbb{P}\left(\bigcap_{n=1}^{\infty} (A \cap C_n)\right)$

[$\{A \cap C_n\}$ is also a decreasing sequence, and $\mathbb{P}$ is a finite measure]

$= \lim_{n \to \infty} \mathbb{P}(A \cap C_n)$

$= \lim_{n \to \infty} \mathbb{P}(A) \cdot \mathbb{P}(C_n)$

$= \lim_{n \to \infty} \mathbb{P}(A) \cdot \mathbb{P}(C_n)$

$= \mathbb{P}(A) \cdot \lim_{n \to \infty} \mathbb{P}(C_n)$

$= \mathbb{P}(A) \cdot \mathbb{P}\left(\bigcap_{n=1}^{\infty} C_n\right)$

$\bigcap_{n=1}^{\infty} C_n \in \mathcal{C}$.

$\therefore$ $\mathcal{C}$ is a monotone class

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$$var(X+Y) = var(X) + var(Y) + 2\,cov(X,Y)$$

$$= var(X) + var(Y) + 2 \times 0$$

$$= var(X) + var(Y)$$

$$cov(X,Y)$$

$$= E(XY) - E(X) \cdot E(Y)$$

$$= E(X) \cdot E(Y) - E(X) \cdot E(Y)$$

$$= 0$$

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Q.6. Suppose that $\mathcal{F}$ is a collection of subsets of Ω with the following properties: $\Omega \in \mathcal{F}$ and $A, B \in \mathcal{F}$ implies $A \setminus B := A \cap B^c \in \mathcal{F}$. Show that $\mathcal{F}$ is a field.

Sol. To show that $\mathcal{F}$ is a field, we have to show that,

① $\phi \in \mathcal{F}$

② If $A \in \mathcal{F}$ then $A^c \in \mathcal{F}$

③ If $A_1, A_2, \ldots, A_n \in \mathcal{F}$, then $\bigcup_{i=1}^{n} A_i \in \mathcal{F}$

First we prove ②.

Let $A \in \mathcal{F}$, $\Omega \in \mathcal{F}$,

then $\Omega \setminus A = \Omega \cap A^c = A^c \in \mathcal{F}$

Now, we prove ①.

By ②, If $A \in \mathcal{F}$, $A^c \in \mathcal{F}$

then, since $\Omega \in \mathcal{F}$, $\Omega^c \in \mathcal{F} \Rightarrow \phi \in \mathcal{F}$

Now, we prove ③,

Let $A_1, A_2, \ldots, A_n \in \mathcal{F}$

then by ②,

$A_1^c, A_2^c, \ldots, A_n^c \in \mathcal{F}$

Now $\bigcup_{i=1}^{n} A_i = \left(\bigcap_{i=1}^{n} A_i^c\right)^c = (A_1^c \cap A_2^c \cap \ldots \cap A_n^c)^c$

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$A_1^c \in \mathcal{F}$, $A_2 \in \mathcal{F} \Rightarrow A_1^c \cap A_2^c \in \mathcal{F}$

$A_1^c \cap A_2^c \in \mathcal{F}$, $A_3 \in \mathcal{F} \Rightarrow A_1^c \cap A_2^c \cap A_3^c \in \mathcal{F}$

Continuing like this,

$A_1^c \cap A_2^c \cap \cdots A_{n-1}^c \in \mathcal{F}$, $A_n \in \mathcal{F}$

$\Rightarrow A_1^c \cap A_2^c \cap \cdots \cap A_{n-1}^c \cap A_n^c = B \in \mathcal{F}$

From ②,

If $B \in \mathcal{F}$, $B^c \in \mathcal{F}$,

then, $(A_1^c \cap A_2^c \cap \cdots \cap A_n^c)^c \in \mathcal{F}$

$\Rightarrow \left(\bigcap_{i=1}^{n} A_i^c\right)^c \in \mathcal{F}$

$\Rightarrow \bigcup_{i=1}^{n} A_i \in \mathcal{F}$

Hence, $\mathcal{F}$ is a field.

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Q7] let $(\Omega_i, \mathcal{F}_i)$, $i=1,2$ be measurable spaces and let, $\mathcal{A}$ be a collection of subset of Ω_2 which generates $\mathcal{F}_2$. Assume that $f: \Omega_1 \to \Omega_2$ satisfies $f^{-1}(A) \in \mathcal{F}_1$, $\forall A \in \mathcal{A}$. Show that f is measurable.

$\Rightarrow$ ~~since~~, let,
$\mathcal{A}$ be a collection of subset of Ω_2 which generates $\mathcal{F}_2$.

i.e. ~~$\sigma(\Omega) = \mathcal{F}_2$~~. $\sigma(\mathcal{A}) = \mathcal{F}_2$

assume that,

$f: (\Omega_1, \mathcal{F}_1) \to (\Omega_2, \mathcal{F}_2)$ satisfies

$f^{-1}(A) \in \mathcal{F}_1$, $\forall A \in \mathcal{A}$.

To show,

$\Rightarrow f^{-1}(A) \in \mathcal{F}_1$, $\forall A \in \sigma(\mathcal{A}) (= \mathcal{F}_2)$

ie. To show, f is measurable.

Now, consider, "~~priority~~ The collection of ~~the~~ Good set,

$\mathcal{C} = \{ A \in \sigma(\mathcal{A}) \mid f^{-1}(A) \in \mathcal{F}_1 \}$ $\longrightarrow$ (*)

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Now, to check, $\mathcal{G}$ is σ-field.

(i) $\phi \in \mathcal{F}_2$.

$\therefore f^{-1}(\phi) = \phi \in \mathcal{F}_1$.

$\therefore f^{-1}(\phi) \in \mathcal{F}_1$ for $\phi \in \mathcal{F}_2$.

Thus, $\phi \in \mathcal{G}$.

(ii)

Let, $A \in \mathcal{G}$ To check, whether, $A^c \in \mathcal{G}$.

$A \in \mathcal{G}$.

$\Rightarrow f^{-1}(A) \in \mathcal{F}_1 \quad \forall \quad A \in \mathcal{F}_2 = \sigma(A)$.

Now $A^c \in \mathcal{F}_2$. $\longrightarrow$ ($\because$ σ-field).

To check, $f^{-1}(A^c) \in \mathcal{F}_1$.

$f^{-1}(A^c) = (f^{-1}(A))^c$

Now, $f^{-1}(A) \in \mathcal{F}_1$

$\Rightarrow (f^{-1}(A))^c \in \mathcal{F}_1$ $\longrightarrow$ ($\because$ $\mathcal{F}_1$ is σ-field)

$\therefore (f^{-1}(A^c)) \in \mathcal{F}_1$.

Thus, $A^c \in \mathcal{F}_2 \Rightarrow f^{-1}(A^c) \in \mathcal{F}_1$.

$\therefore A \in \mathcal{G} \Rightarrow A^c \in \mathcal{G}$

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$\mathcal{G}$ is closed under complementation.

i)

Suppose, $A_1, A_2, \ldots \in \mathcal{G}$

To check whether, $\bigcup_{n=1}^{\infty} A_n \in \mathcal{G}$.

i.e. $f^{-1}\left(\bigcup_{n=1}^{\infty} A_n\right) \in \mathcal{F}_1$.

Now, since, $f^{-1}(A_n) \in \mathcal{F}_1$

$\therefore \bigcup_{n=1}^{\infty} f^{-1}(A_n) \in \mathcal{F}_1$ → $\because \mathcal{F}_1$ is σ-field.

~~Thus,~~ ~~$\bigcup_{n=1}^{\infty} A_n \in \mathcal{G}$~~.

& $f^{-1}\left(\bigcup_{n=1}^{\infty} A_n\right) = \bigcup_{n=1}^{\infty} f^{-1}(A_n) \in \mathcal{F}_1$.

Thus, $f^{-1}\left(\bigcup_{n=1}^{\infty} A_n\right) \in \mathcal{F}_1$

$\Rightarrow \bigcup_{n=1}^{\infty} A_n \in \mathcal{G}$.

Thus, $\mathcal{G}$ is closed under unions.

Thus, $\mathcal{G}$ is σ-field.

Thus, from ⓑ, ⊛,

$\mathcal{A} \subseteq \mathcal{G}$.

$\therefore \sigma(\mathcal{A}) \subseteq \mathcal{G}$

& from ⓐ, ⊛ it directly implies,

$\mathcal{G} \subseteq \sigma(\mathcal{A})$.

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Thus, $\sigma(\mathcal{A}) = \mathcal{G}$.

By (*),

$f^{-1}(A) \in \mathcal{F}_1 \quad \forall \; A \in \mathcal{G} = \sigma(\mathcal{A})$.

$\therefore f^{-1}(A) \in \mathcal{F}_1 \quad \forall \; A \in \mathcal{F}_2$.

Thus, f is measurable.

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8) Any continuous map $f: \mathbb{R}^n \to \mathbb{R}^m$ is Borel measurable.

Let, $g: \mathbb{R}^n \to \mathbb{R}$ be continuous. (10)

Let, $g = (g_1, g_2, \cdots, g_n) \Rightarrow g_i : \Omega \to \mathbb{R} \ \forall i = 1(1)n$

$\because g$ is continuous

$\therefore g_i$'s are continuous $\forall i = 1(1)n$.

$\Rightarrow g_i$'s are Borel measurable $\forall i = 1(1)n$.

Now, $\mathcal{B}_{\mathbb{R}^n} = \sigma\left(\{(a_1, b_1] \times (a_2, b_2] \times \cdots \times (a_n, b_n] : -\infty \le a_i < b_i \le \infty \ \forall i = 1(1)n\}\right)$

$g^{-1}\left((a_1, b_1] \times \cdots \times (a_n, b_n]\right)$

$= \{\omega : g(\omega) = (g_1(\omega), \cdots, g_n(\omega)) \in (a_1, b_1] \times \cdots \times (a_n, b_n]\}$.

$= g_1^{-1}((a_1, b_1]) \cap \cdots \cap g_n^{-1}((a_n, b_n])$

$\in \mathcal{F}$ (where, $g_i : (\Omega, \mathcal{F}) \to (\mathbb{R}, \mathcal{B}_{\mathbb{R}})$

$\Rightarrow g_i^{-1}(\cdot) \in \mathcal{F} \ \forall i = 1(1)n$

$\Rightarrow \bigcap_{i=1}^{n} g_i^{-1}(\cdot) \in \mathcal{F}$)

$\therefore g$ is Borel measurable.

Now, $f: \mathbb{R}^n \to \mathbb{R}^m$ is continuous.

$\Rightarrow f^{-1}(U)$ is open in $\mathbb{R}^n$ whenever U is open in $\mathbb{R}^m$

$\Rightarrow f^{-1}(U) \in \mathcal{B}_{\mathbb{R}^n}$ whenever U is open in $\mathbb{R}^m$

Now, we know, $f^{-1}(A) \in \mathcal{F}_1 \ \forall A \in \mathcal{A}$.

$\Rightarrow f^{-1}(A) \in \mathcal{F}_1 \ \forall A \in \sigma(\mathcal{A})$.

& $\sigma(\{\text{open sets in } \mathbb{R}^m\}) = \mathcal{B}_{\mathbb{R}^m}$.

$\therefore f^{-1}(U) \in \mathcal{B}_{\mathbb{R}^n}$ whenever $U \in \mathcal{B}_{\mathbb{R}^m}$.

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9. a.

Let, $f_1 = \sup f_n$

$\therefore f_1 < a$ if $\forall n, f_n < a$

$\therefore f_1^{-1}(-\infty, a) = \bigcap_{n=1}^{\infty} f_n^{-1}(-\infty, a)$

Now, each f_n is measurable

$\therefore f_n^{-1}(-\infty, a)$ is measurable.

$\therefore$ Their countable intersection $\bigcap_{n=1}^{\infty} f_n^{-1}(-\infty, a)$ is measurable.

$\therefore f_1^{-1}(-\infty, a)$ is measurable.

$\therefore f_1$ is Borel measurable.

Let, $f_2 = \inf f_n = -(\sup(-f_n))$

$\therefore f_2 < b$ if $\exists$ some n such that $f_n < b$

$\therefore f_2^{-1}(-\infty, b) = \bigcup_{n} f_n^{-1}(-\infty, b)$

Now, each f_n is Borel measurable

$\therefore f_n^{-1}(-\infty, b)$ is measurable

$\therefore$ Their countable union $\bigcup_{n=1}^{\infty} f_n^{-1}(-\infty, b)$ is measurable

$\therefore f_2^{-1}(-\infty, b)$ is measurable.

$\therefore f_2$ is Borel measurable.

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9) (b) Let $\{f_n\}$ be a seq^n of $\mathbb{R}$-valued Borel m'ble fn^s defined on Ω. S.T. the functions $\omega \longmapsto \limsup_n f_n(\omega)$ and $\omega \longmapsto \liminf_n f_n(\omega)$ are $\bar{\mathbb{R}}$-valued Borel m'ble fn^s.

Soln:

We know, for any sequence $\{\omega_n\} \subseteq \bar{\mathbb{R}}$, we have

$\sup_{n\geq 1} \omega_n = -\inf_{n\geq 1}(-\omega_n)$ and $\sup_{n\geq 1} \omega_n > c$

iff there exists $i : x_i > c$, $c \in \mathbb{R}$.

Then,

$$\{\omega : \sup_{n\geq 1} f_n(\omega) > c\} = \{\omega : \exists\ i \text{ for which } f_i(\omega) > c\}$$

$$= \bigcup_{i\geq 1} \{\omega : f_i(\omega) > c\} \in \mathcal{F}$$

Since, each f_i is Borel m'ble and Borel σ-field is closed under countable unions; hence,

$\sup_{n\geq 1} f_n$ is Borel m'ble.

And for the infimum, we have,

$\inf_{n\geq 1} f_n = -\sup_{n\geq 1}(-f_n)$ is also Borel m'ble.

Now, $\liminf_{n\to\infty} f_n = \sup_{n\geq 1}\left(\inf_{r\geq n} f_r\right)$

and $\limsup_{n\to\infty} f_n = \inf_{n\geq 1}\left(\sup_{r\geq n} f_r\right)$

P.T.O

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So, obviously,

$\liminf_n f_n(\omega)$ and $\limsup_n f_n(\omega)$ are $\bar{\mathbb{R}}$-valued Borel m'ble functions. (Proved)

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10. let λ denote the Lebesgue measure on $(\mathbb{R}, \mathcal{B}_{\mathbb{R}})$. for any Borel set A and $x \in \mathbb{R}$, define $A+x := \{a+x : a \in A\}$. Show that $A+x$ is again a Borel set and that λ is translation invariant, ie,

$$\lambda(A) = \lambda(A+x), \quad \forall\, x \in \mathbb{R},\ A \in \mathcal{B}_{\mathbb{R}}.$$

If $A \in \mathcal{B}_{\mathbb{R}}$ then shifting all elements $a \in A$ by adding some $x \in \mathbb{R}$ gives us a new set $A+x$. such that,

$$A+x := \{a+x : a \in A\}.$$

Consider a sequence of open intervals $\{I_n\}_{n=1}^{\infty}$ that cover A where a cover of a set is a collection of sets whose union contains A as a subset.

Since $\{I_n\}_{n=1}^{\infty}$ covers A, then $\{I_n + x\}_{n=1}^{\infty}$ is a sequence of open intervals that cover $A+x$.

Here λ is the lebesgue measure.

Thus, $\lambda(A+x) \le \sum_{n=1}^{\infty} l(I_n + x) = \sum_{n=1}^{\infty} l(I_n) \quad (\because l(\{x\}) = 0)$

So for every sequence of open intervals $\{I_n\}_{n=1}^{\infty}$ that cover A, we have, $\lambda(A+x) \le \sum_{n=1}^{\infty} l(I_n)$

$$\Rightarrow \quad \lambda(A+x) \le \lambda(A) \quad \text{---} \quad (1)$$

Now, let $\{I_n\}_{n=1}^{\infty}$ be a seq^n of open intervals that cover $(A+x)$.

Then $\{I_n - x\}_{n=1}^{\infty}$ is a seq^n of open intervals that cover A.

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Therefore, $\lambda(A) \leq \sum_{n=1}^{\infty} l(I_n - x) = \sum_{n=1}^{\infty} l(I_n)$

$\Rightarrow \quad \lambda(A) \leq \lambda(A+x) \quad \text{---} \quad (2)$

From (1) and (2), we conclude that $\lambda(A) = \lambda(A+x)$.

If $A \in \mathcal{B}_{\mathbb{R}}$, To show: $A+x \in \mathcal{B}_{\mathbb{R}}$ $\forall\, x \in \mathbb{R}$.

set of all open sets in $\mathbb{R}$ is a generating set of $\mathcal{B}_{\mathbb{R}}$. So A can be written as:

$A = A_1 \cup A_2 \cup \cdots \; C_1^c \cup C_2^c \cup \cdots$

where $A_i = (a_i, b_i)$ and $C_i = (c_i, d_i)$

By definition of $A+x$,

$A+x = (A_1+x) \cup (A_2+x) \cup \cdots \cup (C_1^c+x) \cup (C_2^c+x) \cup \cdots$

$A_i + x = (a_i+x, b_i+x)$ is an open interval, thus $\in \mathcal{B}_{\mathbb{R}}$

$C_i^c + x = ((-\infty, c_i] \cup [d_i, \infty)) + x$

$= (-\infty, c_i+x] \cup [d_i+x, \infty) \quad \in \mathcal{B}_{\mathbb{R}}$ as any set —

— of the form $[k, \infty), (-\infty, k] \in \mathcal{B}_{\mathbb{R}}$

Thus, $A+x \in \mathcal{B}_{\mathbb{R}}$ by closure under countable unions.

Consider $\tau_x : \mathbb{R} \to \mathbb{R}$, $y \mapsto y - x$. // Continuous, hence mble

Then $A+x = \tau_x^{-1}(A)$ is mble, if A is mble

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Question-11: Let $\{A_n\}$ be a sequence of events in $(\Omega, \mathcal{F}, \mathbb{P})$. Prove Boole's Inequality, i.e

$$\mathbb{P}\left(\bigcup_n A_n\right) \leq \sum_n \mathbb{P}(A_n)$$

Solution: For any events in $A_1, A_2, A_3, \ldots$ in our probability space we have

$$\mathbb{P}\left(\bigcup_i A_i\right) \leq \sum_i \mathbb{P}(A_i)$$

One of the axioms of a probability space is that if $B_1, B_2, B_3, \ldots$ are disjoint subsets of the probability space then,

$$\mathbb{P}\left(\bigcup_i B_i\right) = \sum_i \mathbb{P}(B_i)$$

this is called Countable additivity

Now if $B \subset A$ then $\mathbb{P}(B) \leq \mathbb{P}(A)$

Indeed, from the axioms of a probability distribution,

$$\mathbb{P}(A) = \underbrace{\mathbb{P}(B)}_{\geq 0} + \underbrace{\mathbb{P}(A-B)}_{\geq 0}$$

Now we have to modify the sets A_i, so they become disjoint.

$$B_i = A_i - \bigcup_{j=1}^{i-1} A_j \quad \forall\, i = 1, 2, \ldots$$

So if $B_i \subset A_i$, then we know

$$\bigcup_{i=1}^{\infty} B_i = \bigcup_{i=1}^{\infty} A_i$$

Therefore we can deduce the following equation

$$\mathbb{P}\left(\bigcup_{i=1}^{\infty} A_i\right) = \mathbb{P}\left(\bigcup_i B_i\right) = \sum_i \mathbb{P}(B_i) \leq \sum_i \mathbb{P}(A_i)$$

i.e $$\mathbb{P}\left(\bigcup_n A_n\right) \leq \sum_n \mathbb{P}(A_n)$$

(Proved).

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12) Let μ and ν be two probability measures on $(\mathbb{R}, \mathcal{B}_{\mathbb{R}})$ with the same distribution function. Show that $\mu = \nu$.

→ Given, $\mu((-\infty, x]) = F(x)$ $\forall\, x \in \mathbb{R}$

and $\nu((-\infty, x]) = F(x)$

$$B := \{x \in \mathbb{R} \mid (-\infty, x]\}$$

Consider $M := \{A \in \mathcal{B}_{\mathbb{R}} \mid \mu(A) = \nu(A)\}$ —— (*)

So, $B \subseteq M$

and we know $\sigma(B) = \mathcal{B}_{\mathbb{R}}$

By Monotone class theorem, $\sigma(B) = \mathcal{B}_{\mathbb{R}} \subseteq M$ —— (**)

From (*), $M \subseteq \mathcal{B}_{\mathbb{R}}$ —— (***)

Hence from (**) and (***), $M = \mathcal{B}_{\mathbb{R}}$

Hence $\forall\, A \in \mathcal{B}_{\mathbb{R}}$, $\mu(A) = \nu(A)$.

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Question - [16]

Given, $\{A_n\}$ is a sequence of subsets of Ω.

we are to show, for all $\omega \in \Omega$,

(i) $\liminf_{n\to\infty} 1_{A_n}(\omega) = 1_{\liminf_{n\to\infty} A_n}(\omega)$

L.H.S. $= \liminf_{n\to\infty} 1_{A_n}(\omega) = \sup_{n\geq 1} \inf_{m\geq n} 1_{A_m}(\omega)$

[as, for a real valued sequence, $\{x_n\}_n$

$\liminf_{n\to\infty} x_n = \sup_{n\geq 1} \inf_{m\geq n} x_m$, here, $x_n = 1_{A_n}(\omega)$]

now, $\inf_{m\geq n} 1_{A_m}(\omega) = \begin{cases} 1 & \text{if } \forall\, m \geq n,\ \omega \in A_m \\ 0 & \text{otherwise} \end{cases}$

[As, $1_{A_n}(\omega) = \begin{cases} 1, & \text{if } \omega \in A_n \\ 0, & \text{otherwise} \end{cases}$]

$= \begin{cases} 1 & \text{if } \omega \in \bigcap_{m=n}^{\infty} A_m \\ 0 & \text{otherwise} \end{cases}$

now, $\sup_{n\geq 1} \inf_{m\geq n} 1_{A_n}(\omega) = \begin{cases} 1 & \text{if for atleast one } n, \\ & \omega \in \bigcap_{m=n}^{\infty} A_m \\ 0 & \text{otherwise} \end{cases}$

$= \begin{cases} 1 & \text{if } \omega \in \bigcup_{n=1}^{\infty} \bigcap_{m=n}^{\infty} A_m \\ 0 & \text{otherwise} \end{cases}$

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$$\Rightarrow \liminf_{n\to\infty} 1_{A_n}(\omega) = \begin{cases} 1 & \text{if } \omega \in \bigcup_{n=1}^{\infty}\bigcap_{m=n}^{\infty} A_m \text{ i.e. } \omega \in \liminf_{n\to\infty} A_n \\ 0 & \text{otherwise.} \end{cases}$$

$$= 1_{\liminf_{n\to\infty} A_n}(\omega) \quad \text{(proved)}$$

Now, we are to prove, (ii) $\limsup_{n\to\infty} 1_{A_n}(\omega) = 1_{\limsup_{n\to\infty} A_n}(\omega)$

we know that, $\limsup_{n\to\infty} 1_{A_n}(\omega) = \inf_{n\geq 1} \sup_{m\geq n} 1_{A_n}(\omega)$

now, $\sup_{m\geq n} 1_{A_n}(\omega) = \begin{cases} 1 & \text{if for atleast one } m \geq n,\ \omega \in A_m \\ 0 & \text{otherwise} \end{cases}$

$$= \begin{cases} 1 & \text{if } \omega \in \bigcup_{m=n}^{\infty} A_m \\ 0 & \text{otherwise} \end{cases}$$

now, $\inf_{n\geq 1} \sup_{m\geq n} 1_{A_n}(\omega) = \begin{cases} 1 & \text{if for all } n,\ \omega \in \bigcup_{m=n}^{\infty} A_m \\ 0 & \text{otherwise} \end{cases}$

$$= \begin{cases} 1 & \text{if } \omega \in \bigcap_{n=1}^{\infty}\bigcup_{m=n}^{\infty} A_m \text{ i.e. } \omega \in \limsup_{n\to\infty} A_n \\ 0 & \text{otherwise} \end{cases}$$

$$\therefore \limsup_{n\to\infty} 1_{A_n}(\omega) = \begin{cases} 1 & \text{if } \omega \in \bigcap_{n=1}^{\infty}\bigcup_{m=n}^{\infty} A_m \text{ i.e. } \omega \in \limsup_{n\to\infty} A_n \\ 0 & \text{otherwise} \end{cases}$$

$$= 1_{\limsup_{n\to\infty} A_n}(\omega). \quad \text{(proved)}$$

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17. f, g are borel measurable and integrable functions

and $\int_A f\,d\mu \le \int_A g\,d\mu \quad \forall\ A \in \mathcal{F}$

to prove:

$f \le g \quad \mu$ a.e.

Note: $f, g : (\Omega, \mathcal{F}, \mu) \to (\mathbb{R}, \mathcal{B}_{\mathbb{R}})$

Consider the set
$A := \{\omega : f(\omega) > g(\omega)\}$
If $\mu(A) > 0$, then
$\int (g-f) 1_A \, d\mu < 0$ (Contradiction

we have

$$\int_A g\,d\mu \ge \int_A f\,d\mu$$

$$\Rightarrow \int_\Omega (g-f) 1_A \, d\mu \ge 0 \qquad \text{let } g-f = h$$

case 1. let h be an indicator function 1_B

then $h(x) = \begin{cases} 1 & x \in B \\ 0 & o/w \end{cases}$

thus $h(x) \ge 0 \quad \forall\ x \in \Omega$.

so $h(x) \ge 0 \quad \mu$ a.e.

case 2. let $h = \sum_{j=1}^{\infty} c_j 1_{B_j}$ B_j are disjoint, $B_j \in \mathcal{F}$ $\forall\ j$

and $\bigsqcup_{j=1}^{\infty} B_j = \Omega$

Note: If $\bigsqcup_{j=1}^{m} B_j = \Omega$, we can take $B_{m+1} = B_{m+2} = \cdots = \phi$

Let $A_i \in \mathcal{F}$ such that $h(x) < 0 \quad \forall\ x \in A_i$

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$$h 1_{A_i} = \sum_{j=1}^{\infty} c_j 1_{A_i \cap B_j}$$

Since $A_i \in \mathcal{F} \Rightarrow A_i \subseteq \Omega$, and $\bigsqcup_{j=1}^{\infty} B_j = \Omega$

So there exists atleast one B_{j_K} such that $A_i \cap B_{j_K} \neq \phi$
let $x \in A_i \cap B_{j_K}$

$h(x) 1^{(x)}_{A_i \cap B_{j_K}} = h(x) < 0$ as $h(x) \leq 0 \ \forall \ x \in A_i$ and $x \in A_i \cap B_{j_K}$

$\Rightarrow c_{j_K} 1_{A_i \cap B_{j_K}} < 0$ $\Rightarrow x \in A_i$

$\Rightarrow c_{j_K} < 0$ ——— (i)

Now $\int_{\Omega} h(x) 1^{(x)}_{A_i \cap B_{j_K}} \, d\mu \geq 0$

$\Rightarrow \int_{\Omega} c_{j_K} 1_{A_i \cap B_{j_K}} \, d\mu \geq 0$

$\Rightarrow c_{j_K} \mu(A_i \cap B_{j_K}) \geq 0$ ——— (ii)

$\mu(A_i \cap B_{j_K}) \geq 0$ and $c_{j_K} < 0$ — from (i)

So (iii) is valid only if

$\mu(A_i \cap B_{j_K}) = 0$

This must hold for every B_{j_K} such that $A_i \cap B_{j_K} \neq \phi$

So $\mu\left[\bigsqcup_{k=1} A_i \cap B_{j_K}\right]$

So $\mu\left[\bigsqcup_{k \text{ s.t. } A_i \cap B_{j_K} \neq \phi} A_i \cap B_{j_K}\right] = \sum_{k \text{ s.t. } A_i \cap B_{j_K} \neq \phi} \mu(A_i \cap B_{j_K}) = \sum 0 = 0.$

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~~Thus~~ A_i

Also note that $\bigsqcup_{\substack{k: \\ A_i \cap B_{j_k} \neq \phi}} A_i \cap B_{j_k} = A_i$

Thus $\mu(A_i) = 0$

So $h(x) \geq 0 \quad \mu$ a.e.

iii) Let h be a general measurable function

$h = h^+ - h^-$ where $h^+ \geq 0$ & $h^- \geq 0$ are ~~the~~ defined as

$h^+(x) = \max\{h(x), 0\}$

$h^-(x) = -\min\{h(x), 0\}$

Let $h(x) < 0 \quad \forall \ x \in A_i$

$\Rightarrow h(x) = -h^-(x)$

$$\int_\Omega h 1_{A_i} \, d\mu = \int_\Omega h^+ 1_{A_i} \, d\mu - \int_\Omega h^- 1_{A_i} \, d\mu$$

$$= 0 - \int_\Omega h^- 1_{A_i} \, d\mu \quad \left(\text{as } h^+(x) 1_{A_i}(x) = 0 \cdot 1 = 0 \text{ if } x \in A_i \right.$$

$$\left. = h(x) \cdot 0 = 0 \text{ if } x \notin A_i \right)$$

$$= -\int_\Omega h^- 1_{A_i} \, d\mu \geq 0$$

Now as $h(x) < 0 \quad \forall \ x \in A_i \Rightarrow h^- > 0 \quad \forall \ x \in A_i$

$$\int h^+ 1_{A_i} \, d\mu = \sup \left[0 \leq s \leq h 1_{A_i} \int_\Omega s \, d\mu \right]$$

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Since $\bar{h} 1_{A_i} > 0 \quad \forall x \in A_i$

$\Rightarrow \exists \; s = \sum_{j=1}^{\infty} c_j 1_{A_i \cap B_j} > 0 \quad \forall x \in A_i$

$\Rightarrow c_j > 0 \quad \forall j$ s.t. $A_i \cap B_j \neq \phi$

$\Rightarrow \int_{\Omega} \bar{h} 1_{A_i} d\mu \geq \sum_{k} c_{j_k} \mu(A_i \cap B_{j_k})$

$A_i \cap B_{j_k} \neq \phi$

Note that $c_{j_k} > 0$ and $\mu(A_i \cap B_{j_k}) \geq 0$

$\Rightarrow \int_{\Omega} \bar{h} 1_{A_i} d\mu \geq 0$

$\Rightarrow \int_{\Omega} h 1_{A_i} d\mu = -\int_{\Omega} \bar{h} 1_{A_i} d\mu \leq 0$

equality holds only if $\mu(A_i \cap B_{j_k}) = 0 \quad \forall k$ s.t. $A_i \cap B_{j_k} \neq \phi$

As $\int_{\Omega} h 1_{A_i} d\mu \geq 0 \quad \forall A$

So $\int_{\Omega} h 1_{A_i} d\mu = 0$.

$\Rightarrow \mu\left(\bigcup_k A_i \cap B_{j_k}\right) = 0$

$\Rightarrow \mu(A_i) = 0$.

Thus $h \geq 0$ μ a.e.

Hence proved.

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Question 21: Prove the following version of Markov Inequality. Let $f: \Omega \to \mathbb{R}$ be Borel measurable. Then for any $A \in \mathcal{F}$ and $c > 0$ show that

$$\mu(\{|f| \geq c\} \cap A) \leq \frac{1}{c} \int_A |f| \, d\mu$$

Solution:

Let us define a set

$$A_c = \{x \in A : c \leq |f(x)|\}$$

where $1_{A_c}(\omega) = \begin{cases} 1, & \text{if } \omega \in A_c \\ 0, & \text{o.w.} \end{cases}$

$$\therefore \quad c\,1_{A_c} \leq |f(x)|\,1_{A_c}$$

$$\Rightarrow \int c\,1_{A_c}\, d\mu \leq \int |f(x)|\,1_{A_c}\, d\mu$$

$$\Rightarrow c\mu(A_c) \leq \int |f(x)|\,1_{A_c}\, d\mu$$

$$\Rightarrow c\mu(A_c) \leq \int_{A_c} |f(x)|\, \mu(dx)$$

$$\Rightarrow \mu(A_c) \leq \frac{1}{c} \int_{A_c} |f(x)|\, \mu(dx)$$

$$\Rightarrow \mu(\{|f| \geq c\} \cap A) \leq \frac{1}{c} \int_{A_c} |f(x)|\, \mu(dx) = \frac{1}{c} \int_{A_c} |f|\, d\mu$$

$$\leq \frac{1}{c} \int_A |f|\, d\mu$$

(From construction $A_c \subseteq A \Rightarrow \int_{A_c} |f|\, d\mu \leq \int_A |f|\, d\mu$)

(proved)

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23. Fix $a, b \in \mathbb{R}$ with $a < b$. Suppose that $\exists$ functions $f : (a,b) \times \Omega \to \mathbb{R}$ and $g : \Omega \to \mathbb{R}$ such that

1) $w \in \Omega \mapsto f(t,w)$ is μ-integrable for every fixed $t \in (a,b)$

2) $t \in (a,b) \mapsto f(t,w)$ is continuous for every fixed $w \in \Omega$

3) $g \geq 0$ and μ integrable.

4) $|f(t,w)| \leq g(w) \ \forall \ t, w$.

Then show that the function $h : (a,b) \to \mathbb{R}$ defined by

$$h(t) = \int_{\Omega} f(t,w) \, d\mu(w).$$

is continuous.

we need to prove, as $t_n \to t$, $h(t_n) \to h(t)$ where $\{t_n\} \in (a,b]$.

now, given that, $t \in (a,b) \mapsto f(t,w)$ is continuous for every fixed $w \in \Omega$

$\therefore \{t_n\} \in (a,b] \Rightarrow t_n \to t$

$f(t_n, w) \to f(t,w)$ for fixed $w \in \Omega$.

now,

$$\lim_{n \to \infty} h(t_n) = \lim_{n \to \infty} \int_{\Omega} f(t_n, w) \, d\mu(w) \longrightarrow (1).$$

here, $|f(t,w)| \leq g(w) \ \forall \ t, w$.

$\therefore |f(t_n, w)| \leq g(w) \ \forall \ t, w$ where $g \geq 0$ and μ integrable.

i.e. $\int_{\Omega} g \, d\mu < \infty$.

From (1) $\Rightarrow$

Hence, by DCT,

$$\lim_{n \to \infty} h(t_n) = \int \lim_{n \to \infty} f(t_n, w) \, d\mu(w).$$

$$= \int f(t,w) \, d\mu(w) \quad [\text{since, } f \text{ is continuous}]$$

$$= h(t).$$

Hence, as $t_n \to t$,

then, $h(t_n) \to h(t)$ $\therefore h(t)$ is continuous.

[Proved]

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24. Fix $a, b \in \mathbb{R}$ with $a < b$. Suppose that there exist functions $f : (a,b) \times \Omega \to \mathbb{R}$ and $g : \Omega \to \mathbb{R}$ such that

(1) $\omega \in \Omega \longmapsto f(t,\omega)$ is μ integrable for every fixed $t \in (a,b)$

(2) $t \in (a,b) \mapsto f(t,\omega)$ is differentiable (in t) for every fixed $\omega \in \Omega$.

(3) $g \geq 0$ and μ-integrable.

(4) $\left| \frac{\partial}{\partial t} f(t,\omega) \right| \leq g(\omega) \quad \forall\, t, \omega$

Then show that the function $h : (a,b) \to \mathbb{R}$ defined by $h(t) := \int_\Omega f(t,\omega)\, d\mu$ is differentiable with $h'(t) = \int_\Omega \frac{\partial}{\partial t} f(t,\omega)\, d\mu$

Soln: we need to show $\lim_{k \to 0} \frac{h(t+k,\omega) - h(t,\omega)}{k}$ exists and equals $\int \frac{\partial}{\partial t} f(t,\omega)\, d\mu$

Now, $\frac{h(t+k) - h(t)}{k}$

$= \int \frac{f(t+k,\omega) - f(t,\omega)}{k}\, d\mu$

by (2), $f(t,\omega)$ is diff. hence by MVT

$\frac{f(t+k,\omega) - f(t,\omega)}{k} = f'(t+\theta k, \omega) \quad \theta \in (0,1)$

$= \int f'(t+\theta k, \omega)\, d\mu$

$\therefore \lim_{k \to 0} \frac{h(t+k) - h(t)}{k}$

$= \lim_{k \to 0} \int f'(t+\theta k, \omega)\, d\mu$

$= \int \lim_{k \to 0} f'(t+\theta k, \omega)\, d\mu$ (using (4)).

$= \int \frac{\partial}{\partial t} f(t,\omega)\, d\mu$

QED.

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25) a) (From measures to distribution functions)

Let μ be a Lebesgue-Stieltjes measure on $(\mathbb{R}^2, \mathcal{B}_{\mathbb{R}^2})$.

Define $F: \mathbb{R}^2 \to \mathbb{R}$ by

$$F((x_1, x_2)) := \mu\left(\{(a,b) \in \mathbb{R}^2 : -\infty < a \le x_1, -\infty < b \le x_2\}\right)$$

Show that F is a distribution function on $\mathbb{R}^2$, i.e., it is 'right continuous' and 'non-decreasing'.

→ We are given, $F((x_1, x_2)) := \mu((-\infty, x_1] \times (-\infty, x_2])$

Let $y_1 > x_1$ and $y_2 > x_2$

So we need to show $F(y_1, y_2) > F(x_1, x_2)$ where

~~Now~~ $F((y_1, y_2)) := \mu((-\infty, y_1] \times (-\infty, y_2])$

Now $\{(-\infty, x_1] \times (-\infty, x_2]\} \subseteq \{(-\infty, y_1] \times (-\infty, y_2]\}$

So, $\mu\{(-\infty, x_1] \times (-\infty, x_2]\} \le \mu\{(-\infty, y_1] \times (-\infty, y_2]\}$

i.e. $F(x_1, x_2) \le F(y_1, y_2)$

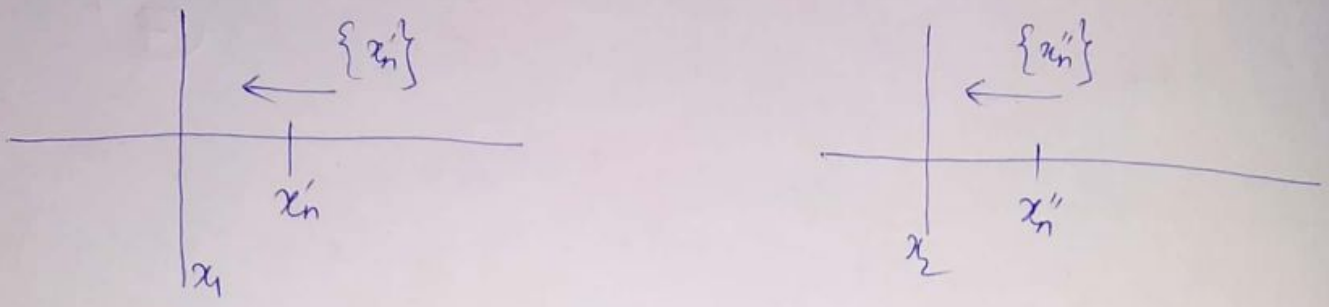
i.e. F is non-decreasing.

Defn given in class.

$F(y_1, y_2) - F(x_1, y_2)$
$- F(y_1, x_2)$
$+ F(x_1, x_2) \ge 0$

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Suppose we have two sequences $\{x_n'\}$ and $\{x_n''\}$ which converge to x_1 and x_2. (from the right)



We need to show $F(x_n', x_n'') \xrightarrow{n\to\infty} F(x_1, x_2)$

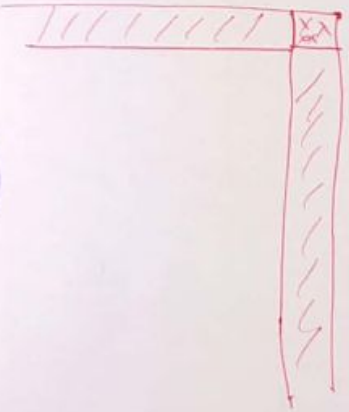
Now $F(x_n', x_n'') := \mu\left((-\infty, x_n'] \times (-\infty, x_n'']\right)$

$$= \mu\left((-\infty, x_1] \times (-\infty, x_2]\right) + \mu\left((x_1, x_n'] \times (x_2, x_n'']\right)$$

There are more terms

$$\lim_{n\to\infty} F(x_n', x_n'') = \mu\left((-\infty, x_1] \times (-\infty, x_2]\right) + \lim_{n\to\infty} \mu\left((x_1, x_n'] \times (x_2, x_n'']\right)$$

$$= F(x_1, x_2)$$



i.e. F is right continuous.

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6) Let $f, g : (\Omega, \mathcal{F}) \to (\mathbb{R}, \mathcal{B}_{\mathbb{R}})$ be measurable functions. Fix real numbers $p, q \in (1, \infty)$ such that $p^{-1} + q^{-1} = 1$. Then, using Young's inequality, prove the following version of Holder's inequality viz.

$$\int_{\Omega} |fg| \, d\mu \leq \left(\int_{\Omega} |f|^p \, d\mu \right)^{1/p} \left(\int_{\Omega} |g|^q \, d\mu \right)^{1/q}$$

$\Rightarrow$ We know, by Young's inequality,

$p, q \in (1, \infty)$ $\ni$ $p^{-1} + q^{-1} = 1$,

then $ab \leq \frac{a^p}{p} + \frac{b^q}{q}$ $\forall a, b \geq 0$.

equality holds if $b = a^{p-1}$

Let us set, $\|f\|_p = \left(\int_{\Omega} |f|^p \, d\mu \right)^{1/p}$, $f \in L^p$

Now, let us take, $a = \frac{|f|}{\|f\|_p}$, $b = \frac{|g|}{\|g\|_q}$

in Young's inequality, we get,

$$\frac{|f|}{\|f\|_p} \frac{|g|}{\|g\|_q} \leq \frac{|f|^p}{p(\|f\|_p)^p} + \frac{|g|^q}{q(\|g\|_q)^q}$$

Integrating both side, we get,

$$\frac{\int_{\Omega} |f||g| \, d\mu}{\|f\|_p \, \|g\|_q} \leq \frac{\int_{\Omega} |f|^p \, d\mu}{p(\|f\|_p)^p} + \frac{\int_{\Omega} |g|^q \, d\mu}{q(\|g\|_q)^q}$$

$$= \frac{(\|f\|_p)^p}{p(\|f\|_p)^p} + \frac{(\|g\|_q)^q}{q(\|g\|_q)^q}$$

$$= \frac{1}{p} + \frac{1}{q}$$

$$= 1$$

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$$\Rightarrow \int_{\Omega} |fg| \, d\mu \leq \|f\|_p \, \|g\|_q$$

$$\Rightarrow \int_{\Omega} |fg| \, d\mu \leq \left(\int_{\Omega} |f|^p \, d\mu \right)^{1/p} \left(\int_{\Omega} |g|^q \, d\mu \right)^{1/q}$$

This is another version of Holder's inequality. [Proved].

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Question-[29]

$\{x_n\}_n$ is a sequence of real numbers.

We know that, $\delta_x(A) = \begin{cases} 1 & \text{if } x \in A \\ 0 & \text{otherwise.} \end{cases}$

We are to prove, $\delta_{x_n} \Rightarrow \delta_0$ iff $x_n \xrightarrow{n\to\infty} 0$

Now, $\delta_{x_n} \Rightarrow \delta_0$ if, $\int_{\mathbb{R}} f(y)\,\delta_{x_n}(dy) \xrightarrow[n\to\infty]{} \int_{\mathbb{R}} f(y)\,\delta_0(dy)$

$\forall$ bounded and continuous $f \overset{n}{\cdot} f : \mathbb{R} \to \mathbb{R}$

Again, we know that, $\int_{\mathbb{R}} f(y)\,\delta_x(dy) = f(x)$

$\therefore \delta_{x_n} \Rightarrow \delta_0$ if, $f(x_n) \xrightarrow{n\to\infty} f(0)$ $\forall$ bounded and continuous $f \overset{n}{\cdot} f : \mathbb{R} \to \mathbb{R}$.

If part:

If $x_n \xrightarrow{n\to\infty} 0$ then, for all bounded continuous $f \overset{n}{\cdot} f : \mathbb{R} \to \mathbb{R}$, $\lim_{n\to\infty} f(x_n) = f(\lim_{n\to\infty} x_n) = f(0)$

[as for a continuous $f \overset{n}{\cdot} f$, $x_n \xrightarrow{n\to\infty} x \Rightarrow f(x_n) \xrightarrow{n\to\infty} f(x)$]

$\therefore f(x_n) \xrightarrow{n\to\infty} f(0)$

$\Rightarrow \delta_{x_n} \Rightarrow \delta_0$ (proved)

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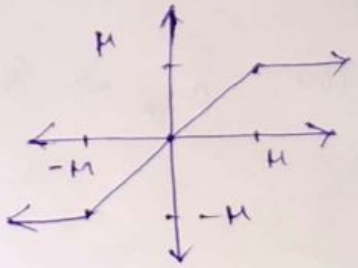
'Only if' part:

We are given that, $\delta_{x_n} \Rightarrow \delta_0$ i.e.

$\forall$ bounded and continuous $f \overset{n}{\in} f : \mathbb{R} \to \mathbb{R}$, $\lim_{n\to\infty} f(x_n) = f(0)$

then we are to prove, $x_n \xrightarrow{n\to\infty} 0$.

Define, $f(x) = \begin{cases} -M & \text{if } x \le -M \\ x & \text{if } |x| \le M \\ M & \text{if } x > M \end{cases}$



then, $f(x)$ is a bounded continuous $f \overset{n}{\in}$.

for, $\{x_n\}$ a sequence of random variable,

we know that, a convergent sequence is necessarily bounded, i.e. if a sequence is not bounded then it is not convergent. Hence, x_n is bounded.

we choose, M such that, $|x_n| \le M \quad \forall\, n$

then, $\lim_{n\to\infty} f(x_n) = f(0)$, for that particular bounded continuous $f \overset{n}{\in} f$,

$\Rightarrow \lim_{n\to\infty} x_n = 0$ (proved)

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20 (a) Let $X, X_1, X_2, X_3, \ldots$ be random variables on some prob. space $(\Omega, \mathcal{F}, P)$. Prove that

$$X_n \xrightarrow[n\to\infty]{\text{in prob}} X \iff E\,\frac{|X_n - X|}{1+|X_n - X|} \xrightarrow{n\to\infty} 0$$

Soln

$X_n \xrightarrow[n\to\infty]{} X$ in prob.

$\Rightarrow X_n - X \xrightarrow[n\to\infty]{} 0$ in prob.

Let $Y_n := X_n - X \;\forall\, n$

for $\varepsilon > 0$,

$$P\{|Y_n| > \varepsilon\} = P\{1+|Y_n| > 1+\varepsilon\}$$

$$= P\left\{\frac{1}{1+|Y_n|} < \frac{1}{1+\varepsilon}\right\}$$

$$= P\left\{1 - \frac{1}{1+|Y_n|} > 1 - \frac{1}{1+\varepsilon}\right\}$$

$$= P\left\{\frac{|Y_n|}{1+|Y_n|} > \frac{\varepsilon}{1+\varepsilon}\right\} \le \frac{E\left(\frac{|Y_n|}{1+|Y_n|}\right)}{\frac{\varepsilon}{1+\varepsilon}}$$

Thus, if $E\,\frac{|Y_n|}{1+|Y_n|} \xrightarrow[n\to\infty]{} 0$ then $P\{|Y_n| > \varepsilon\} \xrightarrow[n\to\infty]{} 0$

$\therefore X_n \xrightarrow[n\to\infty]{} X$ in prob.

$$E\left(\frac{|Y_n|}{1+|Y_n|}\right) = \int \frac{|Y_n|}{1+|Y_n|} f(Y_n)\, dY_n$$

$$= \int_{|Y_n|<\varepsilon} \frac{|Y_n|}{1+|Y_n|} f(Y_n)\, dY_n + \int_{|Y_n|>\varepsilon} \frac{|Y_n|}{1+|Y_n|} f(Y_n)\, dY_n$$

How did you get a density fn? write using P

Now, $|Y_n| < \varepsilon \Rightarrow \frac{|Y_n|}{1+|Y_n|} \le \frac{\varepsilon}{1+\varepsilon}$ also $\frac{|Y_n|}{1+|Y_n|} < 1$

$$\le \frac{\varepsilon}{1+\varepsilon} \int_{|Y_n|<\varepsilon} f(Y_n)\, dY_n + \int_{|Y_n|>\varepsilon} f(Y_n)\, dY_n$$

$$= \frac{\varepsilon}{1+\varepsilon} P(|Y_n| < \varepsilon) + \int_{|Y_n|>\varepsilon} f(Y_n)\, dY_n$$

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$$\leq \frac{\varepsilon}{1+\varepsilon} + P(|Y_n| > \varepsilon)$$

we've $P(|Y_n| > \varepsilon) \to 0$ as $n \to \infty$

$$\therefore \quad \varepsilon \frac{|Y_n|}{1+|Y_n|} \to 0 \text{ as } n \to \infty \quad (\because \varepsilon \text{ is arbitrary})$$

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30(b) $\Rightarrow$ let $X_n \xrightarrow{P} X$ i.e. $\mathbb{P}(|X_n - X| > \varepsilon) \xrightarrow{n\to\infty} 0 \;\forall\, \varepsilon$.

Now if we define

$$A_{n,1} = \{|X_n - X| < 1\}$$

we can say that,

$$E(|X_n - X| \wedge 1) = \int_\Omega (|X_n - X| \wedge 1)\, d\mathbb{P}$$

$$= \int_{A_{n,1}} |X_n - X|\, d\mathbb{P} + \int_{|X_n - X| > 1} 1.\, d\mathbb{P}$$

why? $\leq \int_{A_{n,1}} \varepsilon.\, d\mathbb{P} + \mathbb{P}(|X_n - X| \geq 1)$

$(\mathbb{P}|X_n - X| \geq 1) = 0$ as $n \to \infty$)

$\leq \int_\Omega \varepsilon\, d\mathbb{P} \leq \varepsilon$.

where $\varepsilon \to 0$ as $n \to \infty$.

only if

$E(|X_n - X| \wedge 1) \to 0$

$\Rightarrow |X_n - X| \wedge 1 \xrightarrow{\text{in prob}} 0$ why?

$\Rightarrow |X_n - X| \xrightarrow{\text{in prob}} 0$

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1. Let $X, X_1, X_2, \cdots$ be r.v. on some probability space $(\Omega, \mathcal{F}, \mathbb{P})$

Prove the following equivalences

$$X_n \xrightarrow[n\to\infty]{\text{in Prob.}} X \iff \inf\{\epsilon > 0 : \mathbb{P}(|X_n - X| > \epsilon) < \epsilon\} \xrightarrow{n\to\infty} 0$$

Sol. For r.v. X_n & X, define

$$\eta(X_n, X) = \inf\{\epsilon > 0 : \mathbb{P}(|X_n - X| > \epsilon) < \epsilon\}$$

In the space $\mathcal{M}$ of random variables, form the space $\mathcal{N}$ of equivalence classes

$\{[X_n] : X_n \in \mathcal{M}\}$ where $[X_n]$ is defined as $[X_n] = \{X : X_n = X \text{ a.s.}\}$

We first show that η is well defined on $\mathcal{N}$. To that, let X_n, X_n', X & X' be r.v. with $X_n = X_n'$ a.s. & $X = X'$ a.s.

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It follows that

$$\eta(X_n, X) = \inf\{\epsilon > 0 : \mathbb{P}\{|X_n - X| > \epsilon\} < \epsilon\}$$

$$= \inf\{\epsilon > 0 : \mathbb{P}(|X_n' - X'| > \epsilon) < \epsilon\}$$

$$= \eta(X_n', X')$$

By virtue of this, we will henceforth let a r.v. X_n represents the entire equivalence class $[X_n] = \{X : X_n = X \text{ a.s.}\}$

Let X_n, X & Y be r.v., then

$$\eta(X_n, X) = \inf\{\epsilon > 0 : \mathbb{P}\{|X_n - X| > \epsilon\} < \epsilon\}$$

$\eta(X_n, X) = 0$

if and only if $X_n = X$.

We have next that

$$\eta(X_n, X) = \inf\{\epsilon > 0 : \mathbb{P}\{|X_n - X| > \epsilon\} < \epsilon\}$$

$$= \inf\{\epsilon > 0 : \mathbb{P}\{|X - X_n| > \epsilon\} < \epsilon\}$$

$$= \eta(X, X_n)$$

Finally, we have

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$$\eta(X_n, Y) = \inf\{\epsilon > 0 : \mathbb{P}\{|X_n - Y| > \epsilon\} < \epsilon\}$$

$$= \inf\{\epsilon > 0 : \mathbb{P}\{|X_n - X + X - Y| > \epsilon\} < \epsilon\}$$

$$\leq \inf\{\epsilon > 0 : \mathbb{P}\{|X_n - X| + |X - Y| > \epsilon\} < \epsilon\}$$

$$\leq \inf\{\epsilon > 0 : \mathbb{P}\{|X_n - X| > \epsilon\} < \epsilon\} + \inf\{\epsilon > 0 : \mathbb{P}\{|X - Y| > \epsilon\} < \epsilon\}$$

$$= \eta(X_n, X) + \eta(X, Y)$$

Taken together, we have shown that η is indeed a metric on $\mathcal{N}$. ✓

For the remaining claim, observe that

$X_n \xrightarrow[n\to\infty]{\text{in Prob.}} X$ iff (This is the definition) $\forall\, \epsilon > 0 \quad \lim_{n\to\infty} \mathbb{P}(|X_n - X| > \epsilon) = 0$

The above statements occurs iff, $\exists\, N$

$\exists\,, \forall\, n \geq N$

$$\mathbb{P}\{|X_n - X| > \epsilon\} < \epsilon, \quad \checkmark$$

$$\iff \lim_{n\to\infty} \inf\{\epsilon : \mathbb{P}\{|X_n - X| > \epsilon\} < \epsilon\} = 0$$

$$\iff \eta(X_n, X) \longrightarrow 0 \quad \text{as} \quad n \to \infty$$

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Hence,

$$X_n \xrightarrow[n\to\infty]{\text{in Prob.}} X \iff \lim_{n\to\infty} P\{|X_n - X| > \epsilon\} = 0$$

$$\iff \inf\{\epsilon > 0 : P\{|X_n - X| > \epsilon\} < \epsilon\} \xrightarrow{n\to\infty} 0$$

Hence Proved

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Q.31) Let $X_1, X_2, \ldots$ be R.V on some prob. space $(\Omega, \mathcal{F}, \mathbb{P})$. Suppose $X_n \Rightarrow c$; where $c \in \mathbb{R}$ is some constant. Show that, $X_n \xrightarrow[n\to\infty]{\text{in prob.}} c$.

Ans: $X_n \Rightarrow c$ i.e, X_n converges weakly to c.

Hence, $F_{X_n}(x) \xrightarrow{n\to\infty} F_X(x)$; where

$$F_X(x) = \begin{cases} 0; & x < c; \\ 1; & x \geq c; \end{cases}$$

($F_{X_n}(\cdot) \Rightarrow$ DF of X_n;
$F_X(\cdot) \Rightarrow$ DF of X)

For every $\epsilon > 0$,

$$\mathbb{P}[|X_n - c| < \epsilon] = F_{X_n}((c+\epsilon) - 0) - F_{X_n}(c - \epsilon)$$

Now, $\lim_{n\to\infty} F_{X_n}((c+\epsilon) - 0) = 1$;

While $\lim_{n\to\infty} F_{X_n}(c - \epsilon) = 0$.

$\therefore$ we have, $\mathbb{P}[|X_n - c| < \epsilon] \longrightarrow 1$ as $n \to \infty$.

Hence, $X_n \xrightarrow[n\to\infty]{\text{in prob.}} c$. (Proved)

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22) Let $\{X_n\}$ be a sequence of real valued random variables on a probability space (Ω, F), such the the sequence converges in pth mean to a random variable X. Show that there exists a subsequence $\{X_{n_k}\}$ converging a.s to X.

Solution:

The sequence $\{X_n\}$ converges in pth mean to a r.v X. $\Rightarrow E|X_n - X|^p \xrightarrow{n\to\infty} 0$

Here, for $\forall \epsilon > 0$,

$$P[|X_n - X| > \epsilon] \leq \frac{E|X_n - X|^p}{\epsilon^p}$$

$$\lim_{n\to\infty} P[|X_n - X| > \epsilon] \leq 0$$

$$\Rightarrow \lim_{n\to\infty} P[|X_n - X| > \epsilon] = 0$$

$\Rightarrow X_n$ Converges in Probability to X.

$\{X_n\}$ Converges in pth mean to a r.v X

$\Rightarrow \{X_n\}$ Converges in probability to X.

For any $\epsilon > 0$ $\lim_{n\to\infty} P[|X_n - X| > \epsilon] = 0$

We can observe,

$$\{|X_n - X_m| > \epsilon\} \subseteq \{|X_n - X| > \frac{\epsilon}{2}\} \cup \{|X_m - X| > \epsilon/2\}$$

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$\Rightarrow P\left[|X_n - X_m| > \varepsilon\right] \leq P\left[|X_n - X| > \frac{\varepsilon}{2}\right] + P\left[|X_m - X| > \frac{\varepsilon}{2}\right]$

$\Rightarrow \lim_{n,m \to \infty} P\left[|X_n - X_m| > \varepsilon\right] = 0$

which is cauchy version of convergence in probability.

$\Rightarrow \exists N_\varepsilon$ such that $P\left[|X_n - X_m| > \varepsilon\right] < \varepsilon$.

$\forall n, m \geq N_\varepsilon$.

we are choosing, $\varepsilon = 2^{-K}$

Here we can construct a subsequence $\{X_{n_K}\}$ such that $P\left[|X_{n_K} - X_{n_{K+1}}| > 2^{-K}\right] < 2^{-K}$.

$\forall K = 1, 2, \ldots$

Here, we consider, $A_K = \{|X_{n_K} - X_{n_{K+1}}| > 2^{-K}\}$

Then, $\sum_{K=1}^{\infty} P(A_K) < \infty$ [As $\sum_{K=1}^{\infty} 2^{-K}$ is a convergent series]

So, By Borel-Cantelli lemma,

$P\left[\lim_{K \to \infty} \sup A_K\right] = 0$

$\Rightarrow P\left[A_K \text{ happening infinitely often}\right] = 0$

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if $\omega \in (\limsup A_k)^c = \liminf A_k^c$

events A_k can happen only finitely many times ($\omega \notin A_k$ after large k).

2) $|X_{n_k}(\omega) - X_{n_{k+1}}(\omega)| \leq 2^{-k}$ eventually [for all large k].

Thus $\{X_{n_k}(\omega)\}$ is a cauchy sequence of real numbers. The cauchy property follows since for large l that,

$$\sum_{k \geq l} |X_{n_{k+1}}(\omega) - X_{n_k}(\omega)| \leq \sum_{k \geq l} 2^{-k}$$

$$= 2 \cdot 2^{-l}.$$

therefore for any $r > s$ large, we get

~~$|X_{n_r}(\omega) - X_{n_s}(\omega)|$~~

$$|X_{n_r}(\omega) - X_{n_s}(\omega)| \leq \sum_{k \geq l} |X_{n_{k+1}}(\omega) - X_{n_k}(\omega)|$$

$$\leq 2 \cdot 2^{-l}.$$

This means $\lim_{j \to \infty} X_{n_j}(\omega)$ exists, that is

$\omega \in (\limsup A_k)^c$ implies $\lim_{j \to \infty} X_{n_j}(\omega)$ exists.

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⇒ Therefore for $\omega \in (\limsup A_k)^c$.

& $P(\limsup A_k) = 0$

⇒ $\lim_{k\to\infty} X_{n_k}(\omega) = X(\omega)$.

⇒ $X_{n_k} \xrightarrow{a.s.} X$

⇒ there exists a subsequence $\{X_{n_k}\}$ [which we can construct] converges almost surely to X.

convergence in p-th mean

Markov's ineq ⟹ convergence in probability

done in class ⟹ Subsequences which converge a.s.

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Q.36. Let, $F, F_1, F_2, \ldots$ be the distribution functions corresponding to the probability measures $\mu, \mu_1, \mu_2, \ldots$ on $(\mathbb{R}, \mathcal{B}_{\mathbb{R}})$. Prove that equivalence between the following two statements.

(1) $\lim_{n\to\infty} F_n(x) = F(x)$, for all continuity points x of F, where $F_n(y) := \mu_n((-\infty, y])$, $F(y) := \mu((-\infty, y])$, $\forall y \in \mathbb{R}$ (corresponding distribution functions).

(2) There exists a dense set D in $\mathbb{R}$ such that $\lim_n F_n(x) = F(x)$ $\forall x \in D$.

ans:- (1) $\Rightarrow$ (2)

Let, A be the set of all discontinuity points in F.

~~We know that $\mathbb{R}$ is dense.~~ ~~$\mathbb{R}\setminus A$ is dense in~~

A is countable and

~~Then,~~ $\mathbb{R}\setminus A$ is ~~also~~ dense, and $\mathbb{R}\setminus A$ contains all the points of continuity of F.

Hence, $D = R \setminus A$ is a dense set where $\lim_{n\to\infty} F_n(x) = F(x)$ holds [Proved]

(2) $\Rightarrow$ (1)

~~Fix ϵ~~

Let, $y_1, y_2 \in D \ni y - \epsilon < y_1 < y < y_2 < y + \epsilon$

where $y \in \mathbb{R}$, is a continuity point of F.

and $\epsilon > 0$

It is given that, $\forall x \in D$, $\lim_{n\to\infty} F_n(x) = F(x)$.

Then, we can say that,

$\exists N_1$ such that, $|F_n(y_1) - F(y_1)| \leq \epsilon \quad \forall n \geq N_1$

$\Rightarrow F(y_1) - \epsilon \leq F_n(y_1) \leq F(y_1 + \epsilon)$

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Similarly, $F_{n_2}(y_2) - \epsilon \leq F_n(y_2) \leq F(y_2) + \epsilon \quad \forall\, n \geq N_1$ [we can find such $N_1 \in \mathbb{N}$]

As, both F and F_n's are non-decreasing functions

we can write,

$$F(y-\epsilon) \leq F(y_1) \quad \text{and} \quad F(y_2) \leq F(y+\epsilon)$$

$$\& \quad F_n(y_1) \leq F_n(y) \leq F_n(y_2)$$

Then, $F(y-\epsilon) - \epsilon \leq F(y_1) - \epsilon \leq F_n(y_1) \leq F_n(y) \leq F_n(y_2)$

$$\leq F_{n_2}(y_2) + \epsilon$$

$$\leq F(y+\epsilon) + \epsilon$$

$$\Rightarrow F(y-\epsilon) - \epsilon \leq F_n(y) \leq F(y+\epsilon) + \epsilon$$

Now, Let $n \to \infty$, and then $\epsilon \to 0$,

$$\Rightarrow F(y) \leq \lim_{n\to\infty} F_n(y) \leq F(y)$$

$$\Rightarrow \lim_{n\to\infty} F_n(y) = F(y)$$

Hence the proof.

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1. Let X & Y be independent r.v. defined on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$. Further assume that X & Y are square integrable. Show that X, Y, X^2, Y^2, XY are integrable & that

$$E(XY) = (EX)(EY)$$

$$Var(X+Y) = Var(X) + Var(Y)$$

Sol. X & Y be ind. r.v. on $(\Omega, \mathcal{F}, \mathbb{P})$

$\because$ X & Y are square integrable

$\therefore$ $E[X^2] < \infty$ & $E[Y^2] < \infty$ — (*)

{By defⁿ of square integrable}

Using the inequality

$$|x| \leq 1 + x^2$$

$$|x| \leq \frac{1+x^2}{2}$$

We have $E[|X|] \leq 1 + E[X^2]$

$\because$ $E[X^2] < \infty$

$\therefore$ $E[|X|] \leq 1 + E[X^2] < \infty$

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$\because E[|X|] < \infty$

$\therefore$ By defi^n of integrability of r.v.

i.e Random Variables satisfies

we have a definition of $E|X|$ given in class. $|X|$ could be a mixture of discrete & cont rv

$$E[|X|] = \begin{cases} \sum_{x} |x| p_X(x) < \infty & \text{discrete case} \\ \int_{x} |x| f_X(x)\, d\mu(x) < \infty & \text{cont case} \end{cases}$$

Hence by $E[|X|] \le 1 + E[X^2] < \infty$

shows that a square integrable r.v. is always integrable.

So, X, Y are integrable.

$$E|X^2| = E|X|^2 = EX^2 < \infty$$

$\therefore X^2$ is also integrable [By eq^n ⊛]

$$E|Y^2| = E|Y|^2 = EY^2 < \infty$$

$\therefore Y^2$ is also integrable [By eq^n ⊛]

$E|XY| \le (E|X|^2)^{1/2} (E|Y|^2)^{1/2}$

$|XY| \le |X|\,|Y|$ [By Cauchy-Schwarz Inequality]

$\Rightarrow E|XY| \le E[|X|\,|Y|]$ [Taking expectation]

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$E|XY| \leq E[|X||Y|] = E|X|\,E|Y|$

$\{\because X \& Y$ are ind. $\}$

Arg. is this

$\because X$ & Y are integrable $\left\{ |XY| \leq \frac{X^2+Y^2}{2} \right.$

$E|XY| \leq \frac{EX^2}{2} + \frac{EY^2}{2}$

$\therefore E|X| < \infty$ & $E|Y| < \infty$

$\because EX^2 < \infty$ & $EY^2 < \infty$

$\therefore E|XY| < \infty$

$\Rightarrow XY$ is int.

Hence Proved

So, $E|XY| \leq E|X|\,E|Y| < \infty$

Hence XY is integrable.

(iii) X & Y are independent r.v.

Then by definition of expectation

For $X, Y \geq 0$

$$E(XY) = \int_y \left[\int_x XY \, dF_X(x) \right] dF_Y(y)$$

$\{\because XY$ is integrable $\because E|XY| < \infty\}$

By Fubini's Th^m

$$E(XY) = \int_y \left[\int_x XY \, dF_X(x) \right] dF_Y(y) = \int_x \left[\int_y XY \, dF_Y(y) \right] dF_X(x)$$

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$\because$ X & Y are independent & integrable

$\therefore \quad E[|XY|] = E|X| \; E|Y| < \infty$

Hence, $E(XY) = \int_{\mathcal{X}} \left[\int_{\mathcal{Y}} XY \, dF_Y(y) \right] dF_X(x)$

$$= \int_{\mathcal{X}} x \, dF_X(x) \int_{\mathcal{Y}} y \, dF_Y(y)$$

$$\boxed{E(XY) = (EX)(EY)} \quad \text{— } (*)$$

Hence Proved

By definition of Variance

$$Var(X) = E(X - EX)^2 = EX^2 - (EX)^2$$

$$Var(Y) = E(Y - EY)^2 = EY^2 - (EY)^2$$

$\because$ X & Y are independent & integrable

$$\therefore \quad Var(X+Y) = E(X+Y)^2 - [E(X+Y)]^2$$

$\{\because E|X| < \infty \; \& \; E|Y| < \infty$

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$$Var(X+Y) = E X^2 + E Y^2 + 2E(XY) - (E X + E Y)^2$$

$$= E X^2 + E Y^2 + 2(E X)(E Y) - (E X)^2 - (E Y)^2 - 2(E X)(E Y)$$

By eqⁿ ⊛

$$= E X^2 + E Y^2 - (E X)^2 - (E Y)^2$$

$$= E X^2 - (E X)^2 + E Y^2 - (E Y)^2$$

$$\boxed{Var(X+Y) = Var(X) + Var(Y)}$$

Hence Proved

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38) 1) X & Y are indept iff $\sigma(X)$ & $\sigma(Y)$ are indept.

$\sigma(X) = \{X^{-1}(A) \mid A \in \mathcal{B}_{\mathbb{R}}\}$, $\sigma(Y) = \{Y^{-1}(A) \mid A \in \mathcal{B}_{\mathbb{R}}\}$

$\sigma(X)$ & $\sigma(Y)$ are indept.

$\Leftrightarrow \mathbb{P}(X^{-1}(A) \cap Y^{-1}(B)) = \mathbb{P}(X^{-1}(A))\, \mathbb{P}(Y^{-1}(B)) \quad \forall A, B \in \mathcal{B}_{\mathbb{R}}$.

$\Leftrightarrow \mathbb{P}(X \in A, Y \in B) = \mathbb{P}(X \in A)\, \mathbb{P}(Y \in B) \quad \forall A, B \in \mathcal{B}_{\mathbb{R}}$

$\Leftrightarrow X$ & Y are indept.

2) X & Y are independent

$\Rightarrow \mathbb{P}(X \in A, Y \in B) = \mathbb{P}(X \in A)\, \mathbb{P}(Y \in B) \quad \forall A, B \in \mathcal{B}_{\mathbb{R}}$

Now, $\mathbb{P}(f(X) \in A, g(Y) \in B) \quad \forall A, B \in \mathcal{B}_{\mathbb{R}}$.

$= \mathbb{P}(X \in f^{-1}(A), Y \in g^{-1}(B)) \quad \forall A, B \in \mathcal{B}_{\mathbb{R}}$.

Now, f & g are measurable

$\therefore \forall A, B \in \mathcal{B}_{\mathbb{R}}$, $f^{-1}(A), g^{-1}(B) \in \mathcal{B}_{\mathbb{R}}$

$\therefore \mathbb{P}(X \in f^{-1}(A), Y \in g^{-1}(B)) \quad \forall A, B \in \mathcal{B}_{\mathbb{R}}$

$= \mathbb{P}(X \in f^{-1}(A), Y \in g^{-1}(B)) \quad \forall f^{-1}(A), g^{-1}(B) \in \mathcal{B}_{\mathbb{R}}$

$= \mathbb{P}(X \in f^{-1}(A))\, \mathbb{P}(Y \in g^{-1}(B)) \quad \forall f^{-1}(A), g^{-1}(B) \in \mathcal{B}_{\mathbb{R}}$.

$= \mathbb{P}(f(X) \in A)\, \mathbb{P}(g(Y) \in B) \quad \forall A, B \in \mathcal{B}_{\mathbb{R}}$.

$\therefore f(X)$ & $g(Y)$ are independent if X & Y are independent.

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39). Let X and Y be two real valued, independent r.v defined on $(\Omega, \mathcal{F}, P)$. Suppose that $P(X+Y=a)=1$ for some real no a then show that both X & Y are constant random variable.

$\Rightarrow$ Here $P(X+Y=a)=1$ for some fixed $a \in \mathbb{R}$.

If possible let X is not constant r.v

is $X = \begin{cases} b_1 & \text{w.p } p \\ b_2 & \text{w.p } 1-p. \end{cases}$ why does X only take 2 values?

Then $P(X+Y=a)=1$

$\Rightarrow P(X=b_1, Y=a-b_1)=1$ why?

$\Rightarrow P(X=b_1)\,P(Y=a-b_1)=1$ Since X and Y are independent

$\Rightarrow p\,P(Y=a-b_1)=1$

$\Rightarrow P(Y=a-b_1)=1/p$

Since p is a probability so $0 \le p \le 1$

$\Rightarrow 1 \le 1/p \le \infty$ i $1/p \ge 1$.

which is a contradiction

Hence X can not take more than one value.

Hence X is constant r.v.

Now if $X=b_1$ w.p. 1. Then

$P(Y=a-b_1)=1$. i Y takes value $a-b_1$ w.p 1.

So, Y is also a constant r.v.

Hence X and Y both are constant r.v.

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40) Let X & Y be two real valued, independent, integrable random variable defined on the probability space $(\Omega, \mathcal{F}, P)$. Prove that XY is also integrable.

X & Y are two real valued, independent, integrable rv

X & Y are integrable.

$E|X| < \infty$, $E|Y| < \infty$ ✓

$X = X_+ - X_-$ where, $X_+ = \max\{0, X\}$ and $X_- = \min\{0, X\}$

$Y = Y_+ - Y_-$ where, $Y_+ = \max\{0, Y\}$ and $Y_- = \min\{0, Y\}$

$\therefore X_+, X_-, Y_+, Y_-$ are positive rvs.

X & Y are independent.

we do not know if $E(XY)$ exists.

$E(XY) = E(X)E(Y)$

now, $E|XY| = E[|(X^+ - X^-)(Y^+ - Y^-)|]$.

$= E|X^+Y^+| - E|X^-Y^+| - E|X^+Y^-| + E|X^-Y^-|$

$= E|X^+|E|Y^+| - E|X^-|E|Y^+| - E|X^+|E|Y^-| + E|X^-|E|Y^-|$

$= (E|X^+| - E|X^-|)(E|Y^+| - E|Y^-|)$

$\leq [E(|X^+ - X^-|)][E(|Y^+ - Y^-|)]$ $[\because |a-b| \geq |a| - |b|]$

$\leq E|X| E|Y|$

$< \infty$.

$\therefore E|XY| < \infty$.

$|X|$ & $|Y|$ are independent

Then $E|XY| \leq (E|X|)(E|Y|)$

$< \infty$

$\therefore XY$ is integrable. [Proved].

alternative By cauchy Schwartz's inequality -

$(E|XY|)^2 \leq E|X| E|Y| < \infty$ [as $E|X| < \infty$, $E|Y| < \infty$].

$\Rightarrow E|XY| \leq \infty$.

$\therefore XY$ is integrable [Proved].

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